

EXECUTION PLAN

# Traffy Fondue Community Civic Dashboard

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A community-built transparency platform for Bangkok civic data. *Not another government dashboard. A tool that asks uncomfortable questions.*

Version 1.0 • March 2026 • Bilingual TH/EN

# 1. Positioning — Why This Exists

## The Gap

Traffy Fondue is an incredible platform — 1.37M+ reports, 77% resolution rate, adopted by 17,000+ organizations across Thailand. The official dashboard ([citydata.traffy.in.th](https://citydata.traffy.in.th)) serves its purpose: search tickets, view on a map, download CSV.

But it's a government tool. It's neutral by design. It doesn't ask:

- *Which district has been ignoring flooding complaints for 6 months?*
- *Why does the same pothole on Ratchada Soi 3 get reopened 7 times?*
- *Is my neighborhood getting worse or better over the past year?*
- *Which agency has the worst satisfaction score — and why?*

Your site asks these questions. That's the difference.

## What Already Exists (Competitive Landscape)

| Platform                                                           | What It Does                                                                                        | What It Doesn't Do                                                                     |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Traffy Fondue Dashboard</b>                                     | Map, search, filter, CSV download. Real-time ticket status. Basic stats (total, resolved, pending). | No rankings, no trend analysis, no accountability framing, no neighborhood comparison. |
| <b>BKK Changelog (Creators Garten)</b>                             | Archived ticket history to MongoDB, posted daily changes to Twitter. Had before/after photos.       | Shut down in 2023. Twitter API killed it. No web dashboard survived.                   |
| <b><a href="https://data.bangkok.go.th">data.bangkok.go.th</a></b> | Official BMA open data portal. Has Traffy stats by category (top 25 complaint types).               | Static datasets, no interactive exploration, no community angle.                       |
| <b>Academic research (Frontiers, Chula)</b>                        | Published papers analyzing Traffy data for urban planning, walkability, traffic forecasting.        | Papers, not tools. Not accessible to regular citizens.                                 |

**Your opportunity:** BKK Changelog proved there's community appetite for this. It died because of a dependency on Twitter. You build on open web, you own the platform, and you go deeper than a changelog — you build an accountability layer.

## 2. Product Definition

### Core Identity

|                         |                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Working Name</b>     | เมืองโปร่งใส (Mueang Prong Sai) — “Transparent City”. Or pick your own name. The point is: it’s not “Traffy Dashboard v2.” It has its own identity.   |
| <b>One-Liner (TH)</b>   | เพราะเมืองที่ดี ต้องเริ่มจากการตั้งคำถามที่ถูกต้อง (A better city starts with asking the right questions)                                             |
| <b>One-Liner (EN)</b>   | Community-driven civic transparency for Bangkok. Powered by open data, built for accountability.                                                      |
| <b>Audience</b>         | Bangkok residents who care about their neighborhood. Journalists. Civic hackers. Community leaders. NGOs. Researchers who need clean, queryable data. |
| <b>NOT the audience</b> | Government officials who already have internal tools. Traffy Fondue’s own team (don’t compete, complement).                                           |

### Five Core Pages

*The site has exactly five pages. No feature creep. Each page answers one clear question.*

#### ① Overview / ภาพรวม

**Question it answers:** “How is Bangkok doing right now?”

- Top-level KPIs: total tickets, resolution rate, median fix time, satisfaction rate, reopen rate
- Monthly trend chart (resolved vs pending, 12-month rolling)
- Top 5 problem types by volume with resolution speed per type
- A single provocative stat at the top that changes weekly (e.g., “47 tickets in your area are older than 6 months”)
- *Language toggle TH/EN lives in the header, persists via localStorage*

#### ② District Report Card / ผลงานเขต

**Question it answers:** “How is MY neighborhood doing?”

- Dropdown or clickable map to select one of Bangkok’s 50 districts (เขต)
- Report card format: letter grade (A-F) based on composite score (speed + satisfaction + reopen rate)
- Comparison panel: “Your district vs Bangkok average”
- Top 3 problem types in this district, with trend arrows (improving / worsening)

- Worst unresolved tickets list (oldest open tickets in this district)
- *This is the page people will share on social media. Make it screenshot-friendly.*

### ③ Agency Leaderboard / จัดอันดับหน่วยงาน

**Question it answers:** “Who’s doing well? Who’s not?”

- Sortable table of all organizations from the ‘organization’ field
- Columns: tickets handled, median resolution time, satisfaction score, reopen rate
- Sortable + filterable by problem type
- Green/red indicators for above/below Bangkok average
- *This is the accountability page. Frame it fairly — show both good and bad performers.*

### ④ Before / After Gallery / ก่อน-หลัง

**Question it answers:** “Does reporting actually make a difference?”

- Masonry grid of before/after photo pairs from ‘photo’ and ‘photo\_after’ fields
- Only show tickets where both photos exist AND state = เสร็จสิ้น
- Filterable by type, district
- Each card shows: problem type, district, time to resolve, satisfaction star
- *This is the HOPE page. Motivation for people to keep reporting. Visual proof that the system works.*

### ⑤ Explore Map / แผนที่

**Question it answers:** “What’s happening around me?”

- Leaflet map with clustering (yes, similar to Traffy, but with YOUR filters)
- Unique filter: “Stale tickets only” (open > 90 days), “Low satisfaction only” (star ≤ 2)
- Choropleth mode: color districts by performance metric (speed, volume, satisfaction)
- Click a pin → see full ticket detail including before/after, timeline, organization chain
- *Don’t try to out-map Traffy. Your advantage is the FILTERS that tell a story.*

## 3. Data Architecture

### Data Source

Traffy Fondue's public API provides GeoJSON and CSV exports. Key endpoints:

- **GeoJSON API:** `publicapi.traffy.in.th/teamchadchart-stat-api/geojson/v1` — returns up to 10 features per call with full ticket data
- **Monthly CSV dumps:** Available on `traffy.in.th` open data page, one CSV per month from 2022 onwards
- **Kaggle dataset:** Full historical dataset 2022–2025 available (community-maintained)
- **data.bangkok.go.th:** Official BMA portal has aggregated stats by category

### Recommended Data Pipeline

**Don't try to call the API live from the browser.** The API is rate-limited and returns max 10 results per call. You need a backend data pipeline.

#### Option A: Static Site + Pre-computed Data (Recommended for v1)

- Download monthly CSVs + Kaggle historical dump
- Python script (pandas) to clean, aggregate, compute metrics per district/org/type/month
- Output: JSON files (one per page's data needs) deployed as static assets
- Update cadence: weekly or monthly batch job (GitHub Actions cron)
- **Pros:** Zero hosting cost (Vercel/Netlify free tier), fast, simple, no backend to maintain
- **Cons:** Not real-time, depends on CSV availability

#### Option B: Backend + Database (For v2+)

- Daily cron job polls Traffy API for recently-updated tickets
- Store in PostgreSQL or Supabase (free tier)
- API layer serves pre-aggregated views to the frontend
- **Pros:** Near real-time, can support search, richer queries
- **Cons:** Hosting cost, maintenance, more complex

### Data Processing Steps (for Option A)

- **1. Ingest:** Combine all monthly CSVs into one dataframe
- **2. Clean:** Parse coords into lat/lon, convert timestamps, handle missing values
- **3. Enrich:** Compute derived metrics — `resolution_days`, `is_stale` (open > 90 days), `satisfaction_bucket`

- **4. Aggregate:** Group by district, organization, type, month → compute medians, counts, rates
- **5. Export:** JSON files: overview.json, districts.json, orgs.json, monthly\_trends.json, gallery.json (sample of before/after pairs)

## 4. Tech Stack

| Layer                | Choice                    | Why                                                                                                  |
|----------------------|---------------------------|------------------------------------------------------------------------------------------------------|
| <b>Framework</b>     | Next.js 14+ (App Router)  | Static export for v1, easy upgrade to server components later. Great i18n support.                   |
| <b>Styling</b>       | Tailwind CSS              | Fast iteration, responsive by default. Pair with a custom design system (not default Tailwind look). |
| <b>Charts</b>        | Recharts or Nivo          | React-native charting. Nivo for more polish, Recharts for simplicity. Both support animations.       |
| <b>Maps</b>          | Leaflet via react-leaflet | Free, well-documented, same tech Traffy uses. Supports clustering, choropleth, custom popups.        |
| <b>i18n</b>          | next-intl                 | Handles TH/EN routing, date formatting, pluralization. Proven in production.                         |
| <b>Data pipeline</b> | Python (pandas)           | You already know this stack. Jupyter for exploration, script for production pipeline.                |
| <b>Hosting</b>       | Vercel (free tier)        | Static export = free hosting. Custom domain. Auto-deploy from GitHub.                                |
| <b>CI/CD</b>         | GitHub Actions            | Weekly cron job to re-process CSVs and rebuild. Free for public repos.                               |

### Design Direction (Not Generic)

The design should feel like a journalism/editorial tool, not a SaaS dashboard. Think: The Pudding, Reuters Graphics, or Bangkok Post data stories. Specific principles:

- **Typography:** Noto Sans Thai for Thai text (it's the only font that handles Thai well at all weights). Pair with a strong Latin display font like Instrument Sans, Geist, or Syne for headings.
- **Color:** Dark mode default (city data looks better on dark). Accent color: teal/cyan family. Red for bad metrics, green for good. Amber for caution. Avoid purple gradients.
- **Layout:** Editorial grid. Generous whitespace. Cards with subtle borders, not drop shadows. Data should breathe.
- **Motion:** Framer Motion for page transitions, chart animations, number counters. Smooth scroll between sections. Nothing flashy — purposeful movement only.
- **Mobile:** Mobile-first. 60%+ of Bangkok residents will access this from a phone. Cards stack vertically. Map is full-width. Charts are swipeable.

## 5. Data Column → Feature Mapping

Every column in the Traffy dataset maps to a specific feature. Nothing is wasted.

| Column                                          | How You Use It                                                            | Which Page                      |
|-------------------------------------------------|---------------------------------------------------------------------------|---------------------------------|
| <b>ticket_id</b>                                | Unique reference for deep-link to individual tickets                      | All                             |
| <b>type, problemtype_tag</b>                    | Category filters, breakdown charts, type-specific analysis                | Overview, District, Leaderboard |
| <b>organization, organization_action</b>        | Agency leaderboard rankings, responsibility tracking                      | Leaderboard                     |
| <b>comment</b>                                  | Search functionality (future), NLP sentiment analysis (future)            | Future feature                  |
| <b>coords</b>                                   | Map pin placement, spatial clustering, heatmap density                    | Map                             |
| <b>photo, photo_after</b>                       | Before/after gallery, visual proof of resolution                          | Gallery                         |
| <b>district, subdistrict, province</b>          | District report cards, geographic filtering, comparison                   | District, Map                   |
| <b>address</b>                                  | Detailed location display in ticket detail view                           | Map (popup)                     |
| <b>timestamp</b>                                | Time-series trends, seasonal patterns, volume over time                   | Overview, Trends                |
| <b>state</b>                                    | Resolution rate calculation, stale ticket identification                  | All pages                       |
| <b>star</b>                                     | Satisfaction scoring, quality rankings                                    | Overview, District, Leaderboard |
| <b>count_reopen</b>                             | Quality signal, “repeat offender” identification                          | District, Leaderboard           |
| <b>duration_minutes_*</b>                       | Speed metrics, district/org ranking by response time                      | Overview, District, Leaderboard |
| <b>timestamp_inprogress, timestamp_finished</b> | SLA analysis, time-of-day patterns, weekday vs weekend                    | Trends                          |
| <b>last_activity</b>                            | Stale ticket detection (last_activity far in the past + state ≠เสร็จสิ้น) | District, Map                   |



## 6. Bilingual Strategy (TH/EN)

This is not just a translate button. Thai and English audiences have different needs.

### Implementation

- **URL structure:** /th/overview and /en/overview — clean URLs, SEO-friendly
- **Default language:** Thai. Detect from browser Accept-Language header, remember in localStorage.
- **Toggle:** Persistent toggle in header. Flags or “TH | EN” text. Changes URL and all content.
- **Data labels:** District names, organization names stay in Thai always (they’re proper nouns). Metric labels, UI text, explanations switch.

### Content Strategy Per Language

| Element           | Thai                               | English                                         |
|-------------------|------------------------------------|-------------------------------------------------|
| District names    | เขตพระนคร (native)                 | Phra Nakhon (romanized)                         |
| Metric labels     | เวลาแก้ไขเฉลี่ย                    | Median Resolution Time                          |
| Explanatory text  | Written for residents, casual tone | Written for researchers/expats, more analytical |
| Number formatting | 1,250,410 (same)                   | 1,250,410 (same)                                |
| Date formatting   | 25 มี.ค. 2569 (Buddhist Era)       | Mar 25, 2026 (CE)                               |

## 7. Phased Execution Plan

*Three phases. Ship early, iterate. Don't build everything at once.*

### Phase 1: Foundation (Weeks 1–4)

**Goal:** A live site with the Overview page and District Report Card, populated with real data.

#### Week 1 — Data Pipeline

- Download Kaggle dataset + latest monthly CSVs
- Build Python processing script: clean, enrich, aggregate
- Output: overview.json, districts.json, monthly\_trends.json
- Test with Jupyter notebook: validate all numbers make sense

#### Week 2 — Project Scaffold + Overview Page

- Next.js project setup with next-intl, Tailwind, Framer Motion
- Design system: colors, typography, spacing tokens
- Build Overview page with real data from JSON
- TH/EN toggle working end-to-end

#### Week 3 — District Report Card

- District selector (dropdown + simple map click)
- Report card layout with letter grades
- Comparison: “your district vs Bangkok average”
- Worst unresolved tickets list

#### Week 4 — Polish + Deploy

- Mobile responsive pass (test on actual Thai phones)
- SEO: meta tags, Open Graph images for social sharing
- Deploy to Vercel with custom domain
- GitHub Actions cron for weekly data refresh
- **SHIP IT. Two pages is enough for launch.**

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### Phase 2: Depth (Weeks 5–8)

**Goal:** Add the Leaderboard, Gallery, and Map. The site becomes a complete tool.

### Week 5–6 — Agency Leaderboard + Gallery

- Sortable/filterable leaderboard table
- Before/after photo gallery with lazy loading
- Gallery filters: type, district

### Week 7–8 — Map + Search

- Leaflet map with clustering
- Choropleth mode (district shading by metric)
- Story-driven filters: “stale only”, “low satisfaction”
- Ticket detail popup/panel

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## Phase 3: Community (Weeks 9–12+)

**Goal:** Make the site a platform, not just a dashboard.

- **Shareable report cards:** Generate OG images per district so people share on social media
- **Monthly digest:** Auto-generated summary (“Bangkok Civic Report — March 2026”)
- **API for others:** Expose your cleaned/aggregated data as a public API for other civic hackers
- **Notification system:** LINE or email alerts when a watched district gets a new report card grade
- **Community contributions:** Open source the code, accept PRs, credit Traffy Fondue prominently

## 8. Constraints & Risks

| Risk                                  | Impact                                                                | Mitigation                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>API rate limits / availability</b> | Can't get data fast enough, or API goes down                          | Use monthly CSV dumps as primary source. Don't depend on real-time API.                       |
| <b>Data accuracy</b>                  | Traffy data has gaps: missing coords, duplicate tickets, test entries | Build validation in pipeline. Flag/exclude suspicious data. Be transparent about methodology. |
| <b>Traffy team relations</b>          | They might not like a "wall of shame" built on their data             | Credit them prominently. Frame as "complementary" not "critical." Contact them early.         |
| <b>Political sensitivity</b>          | Rankings can be politically charged (BMA is politically managed)      | Let data speak. No editorializing. Show methodology. Offer raw data download.                 |
| <b>Scope creep</b>                    | Trying to build everything at once                                    | Phase 1 = 2 pages only. Ship in 4 weeks. Iterate based on real feedback.                      |
| <b>Photo hosting costs</b>            | Before/after photos are on Google Storage (Traffy's bucket)           | Hotlink to their URLs. If they block, cache thumbnails only.                                  |

## 9. Credit & Ethics

Traffy Fondue explicitly supports open data reuse. Their page states: if you use their data, credit them and inform their team. Follow this.

- **Always display:** "Data source: Traffy Fondue (traffy.in.th)" with their logo in the footer
- **Link back:** Every ticket detail links to the original on Traffy
- **Inform the team:** Email/contact Traffy before launch. Show them what you've built. They may even feature you.
- **Privacy:** Traffy's data is already public. But if photos contain faces, consider blurring (BKK Changelog used Azure Face API for this).
- **Open source:** Make your code open. Credit everyone who contributes. This is a community project.

**Next step:** Download the Kaggle dataset, run the first data exploration in a Jupyter notebook, and see what the actual numbers look like. Every decision above should be validated against real data before you build.

Build the data pipeline first. The UI comes second. — *Good data makes a mediocre UI useful. Good UI with bad data is useless.*